

PORTFOLIO · 2023–2026

JOHN NORTHRUP

Architectural designer who draws it, builds it, and automates it.

Permit drawing sets for city and county approval, fabrication from concept to install, and Rhino, Grasshopper, and Python tools that speed up design work.

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CODE & AI

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ARCHITECTURE

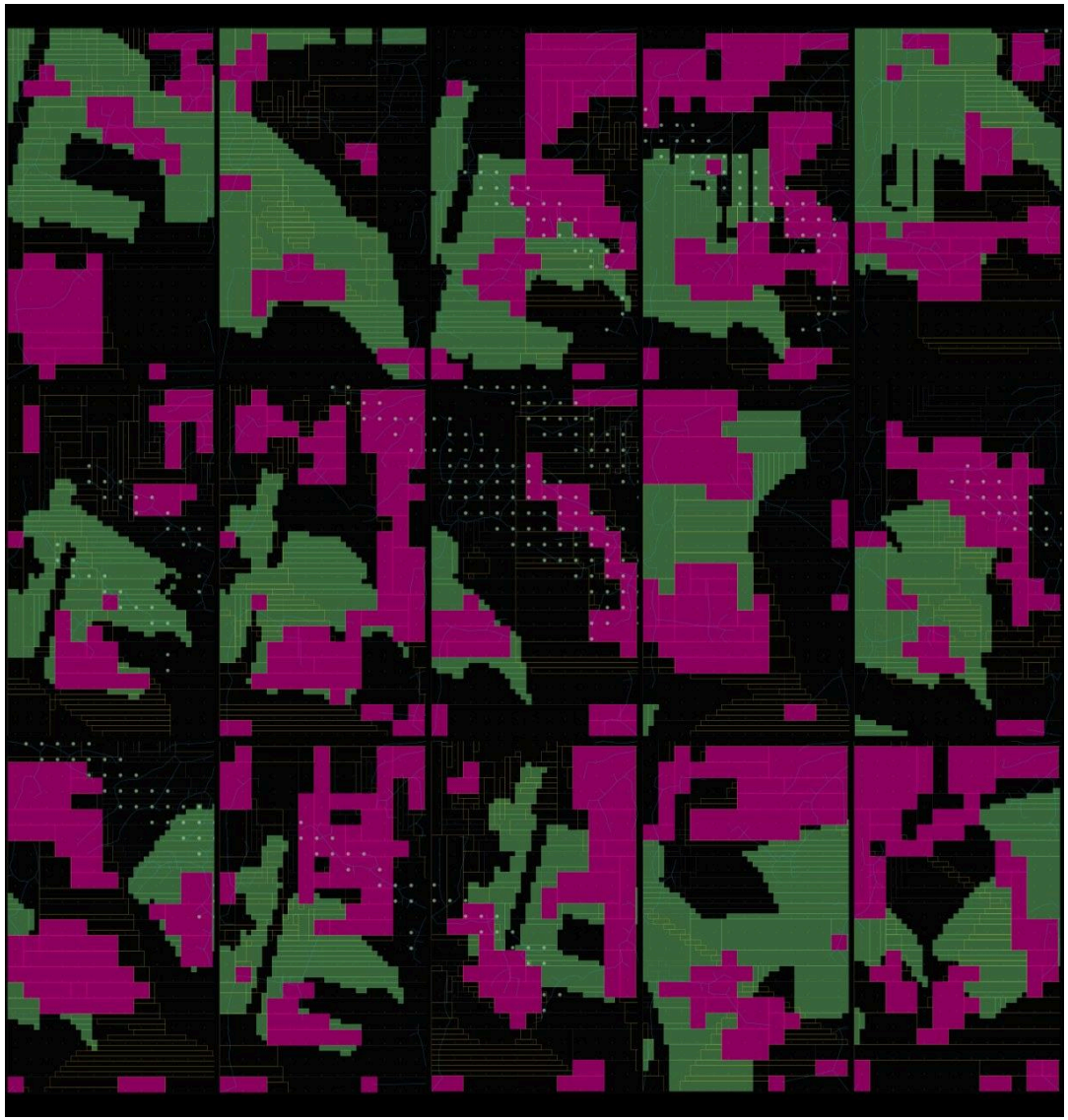
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01

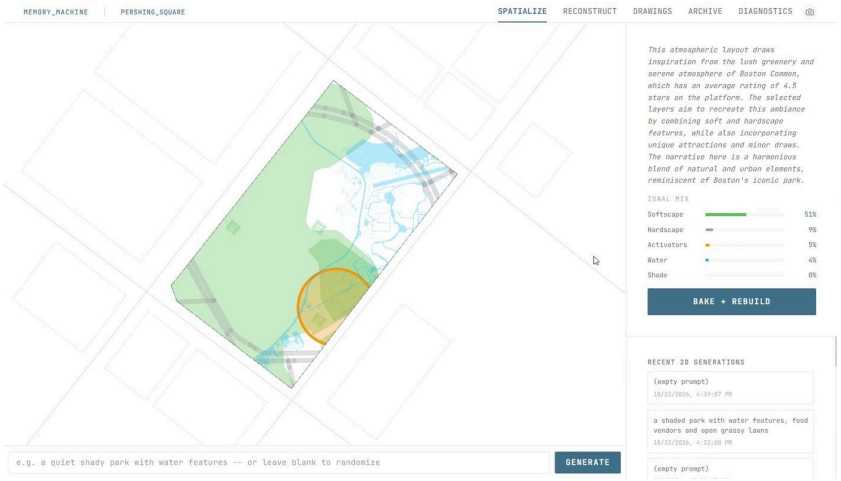
memoryMachine

SU26, M.ARCH THESIS
ADVISOR · CASEY REHM

memoryMachine is a computational design framework that transforms civic sentiment and archival memory into operative spatial data to revitalize under-utilized public space. Using Pershing Square in Downtown Los Angeles as a test bed—a civic site defined by a history of top-down, disconnected redesigns—the project proposes an adaptive spatial infrastructure in place of the fixed, one-time master plan.

At its core, a multimodal data ingestion engine aggregates web reviews, archival records, and spatial metrics, pairing qualitative sentiment with physical variables such as microclimate, canopy coverage, and circulation flows. To ensure equitable representation, the engine applies demographic weighting to mitigate digital access bias. These inputs are translated into spatial density fields and programmatic adjacencies, generating interventions—such as terraced landforms, water systems, and subterranean transit connections—as evidence-based spatial responses rather than predetermined formal gestures.

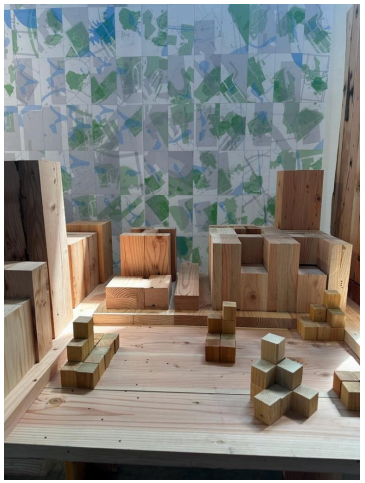
By merging expert analysis with experiential civic data, memoryMachine repositions the architect from sole author to system mediator. Public input shifts from a one-off consultation phase into a permanent, operative design parameter. The result is a self-regulating spatial framework that ingests usage patterns to continuously re-evaluate the site and adapt through phased physical interventions—establishing a democratic model in which public space evolves alongside the community it serves.



SOFTWARE | GENERATE A PLAN, REBUILD IT AS MASSING, DRAW IT, CHECK THE NUMBERS (SP...



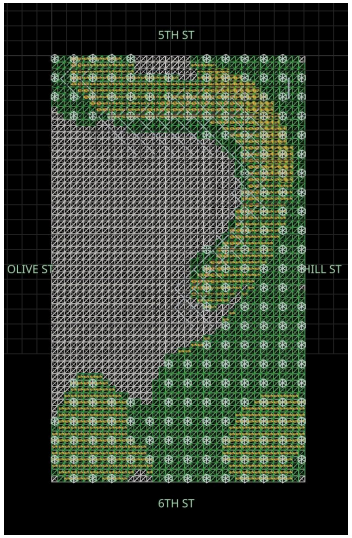
INSTALLATION | TIMBER SITE MODEL IN THE EXHIBITION HALL



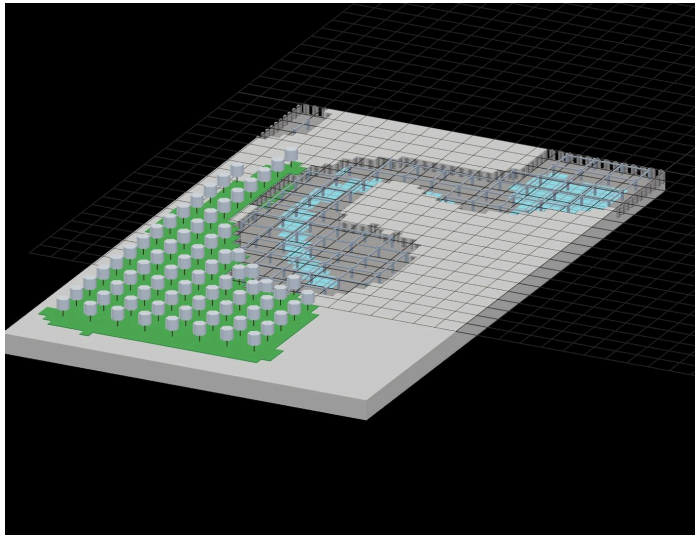
INSTALLATION | TIMBER SITE MODEL A...



RENDER | THE GENERATED PARK IN ITS DOWNTOWN CONTEXT



GENERATED PLAN | PERSHING SQUAR...



GENERATED MODEL | THE SAME SCHEME AS A MASSING



DRAWING | PRECEDENT LIBRARY DRAWN BY THE SYSTEM: PARC DE LA VILLETTE



02

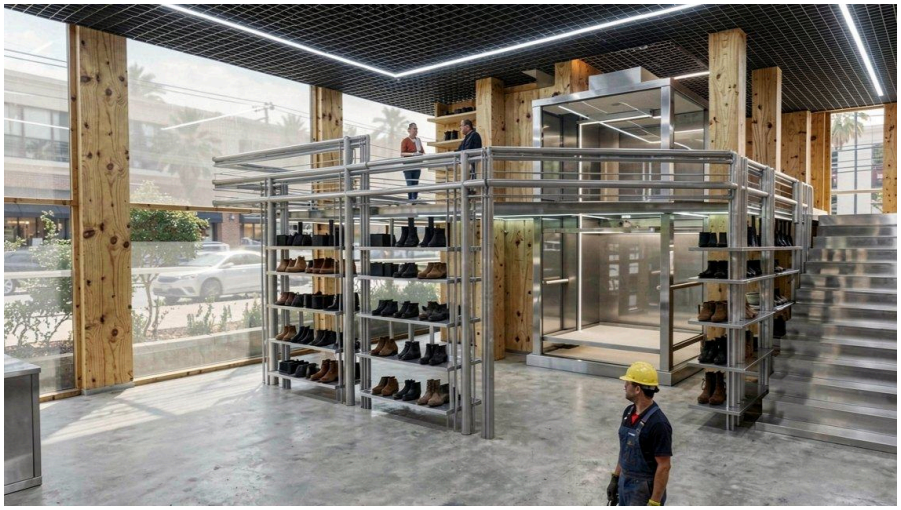
PROTOTYPE

SP26, PROTOTYPE, 3GB STUDIO

A generative pipeline for retail architecture and product design, built around a footwear brand's flagship storefront. Architectural iterations were produced through a custom Flux/ComfyUI pipeline, while the product line — a technical boot shown here in cutaway and colorway studies — was developed in parallel through a Gemini-based image pipeline. The two threads feed each other: the building's industrial material palette of raw wood, steel, and concrete is carried directly into the product's own construction and display.



CUTAWAY SECTION ELEVATION OF THE FULL BUILDING



INTERIOR RETAIL FLOOR WITH STEEL AND TIMBER SHELVING DISPLAY



INTERIOR COUNTER SCENE WITH STAFF AND CUSTOMER



TECHNICAL BOOT PRODUCT RENDER, STUDIO SHOT



BOOT COLORWAY COMPARISON SHEET, FOUR VARIANTS



TECHNICAL CUTAWAY ILLUSTRATION OF THE BOOT'S INTERNAL CO...

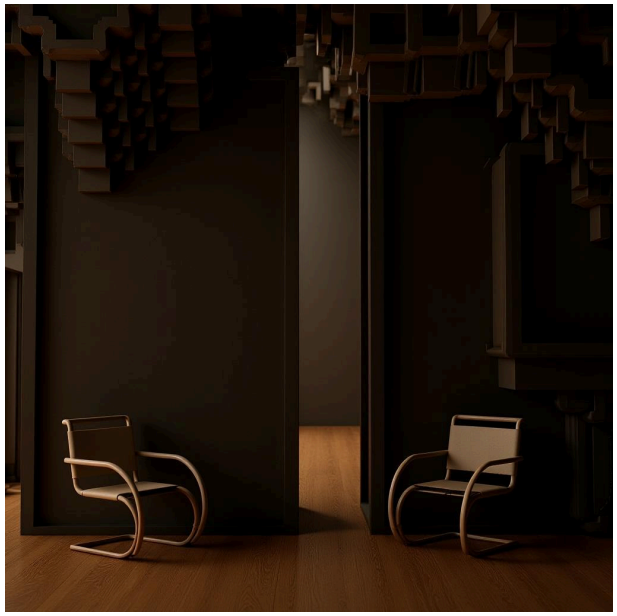


03

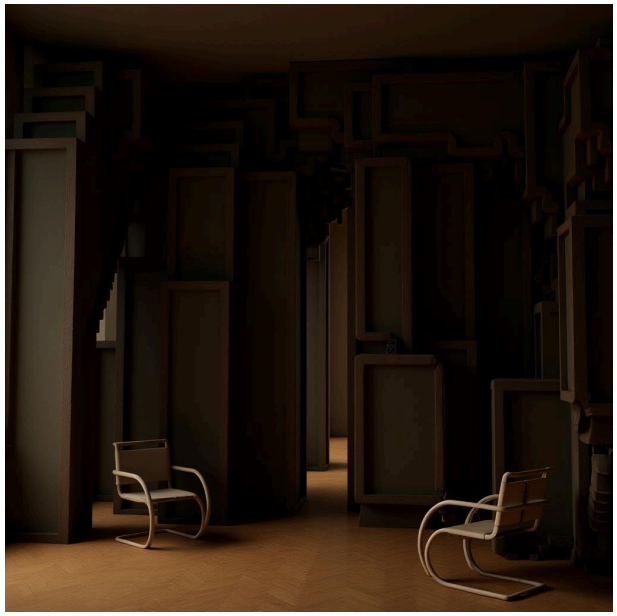
INTERFACE ARCHITECTURE

3GA STUDIO
ADVISOR · CASEY REHM

A Cabinet of Curiosities that produces memory artifacts. The installation receives audio input from the surrounding space; language becomes image through translation of an LLM to 3D object creation via API calls (Meshy). Objects are automatically optimized for 3D printing — oriented, scaled, and combined through a Python pipeline.



TWO SYMMETRIC DARK PORTALS, EACH WITH A SINGLE CHAIR



HALLWAY OF PIXELATED CORNICE PEDESTALS WITH TWO CHAIRS O...



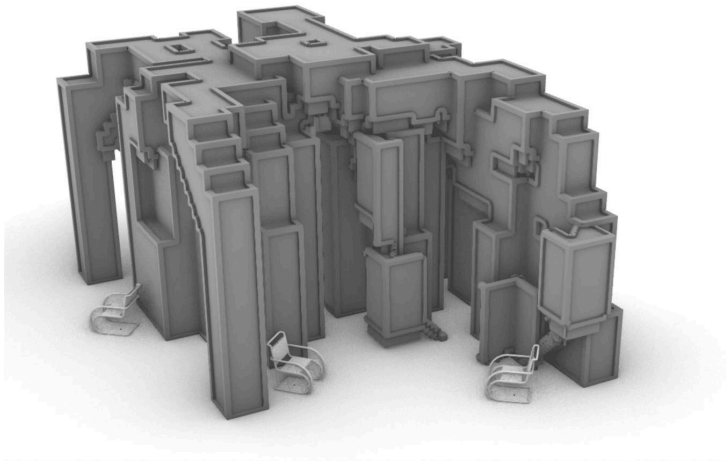
SINGLE CHAIR AGAINST A PIXELATED WOOD CORNICE WALL, DOOR...



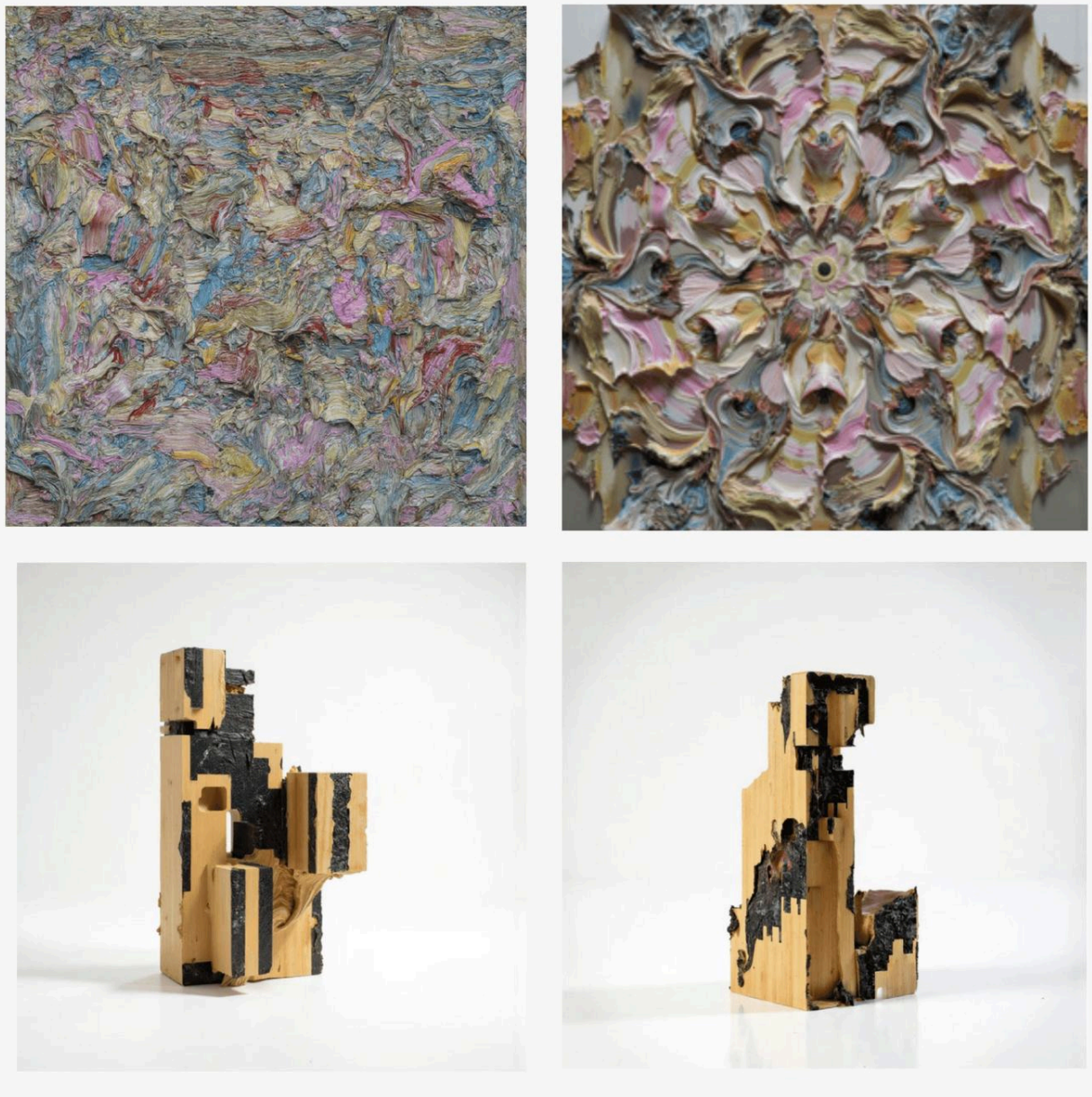
CABINET OF CURIOSITIES INSTALLATION, CROPPED DETAIL VIEW



ELEVATION DRAWING RENDER OF THE CABINET



ASSEMBLED 3D-PRINTED OBJECT RENDER, MASSING VIEW



TEXTURED 3D-PRINTED SCULPTURES, TWO VIEWS

OBJECT STUDIES



BURNT WOOD MEMORY ARTIFACT, STUDY 1



BURNT WOOD MEMORY ARTIFACT, STUDY 2



BURNT WOOD MEMORY ARTIFACT, STUDY 3



BURNT WOOD MEMORY ARTIFACT, STUDY 4

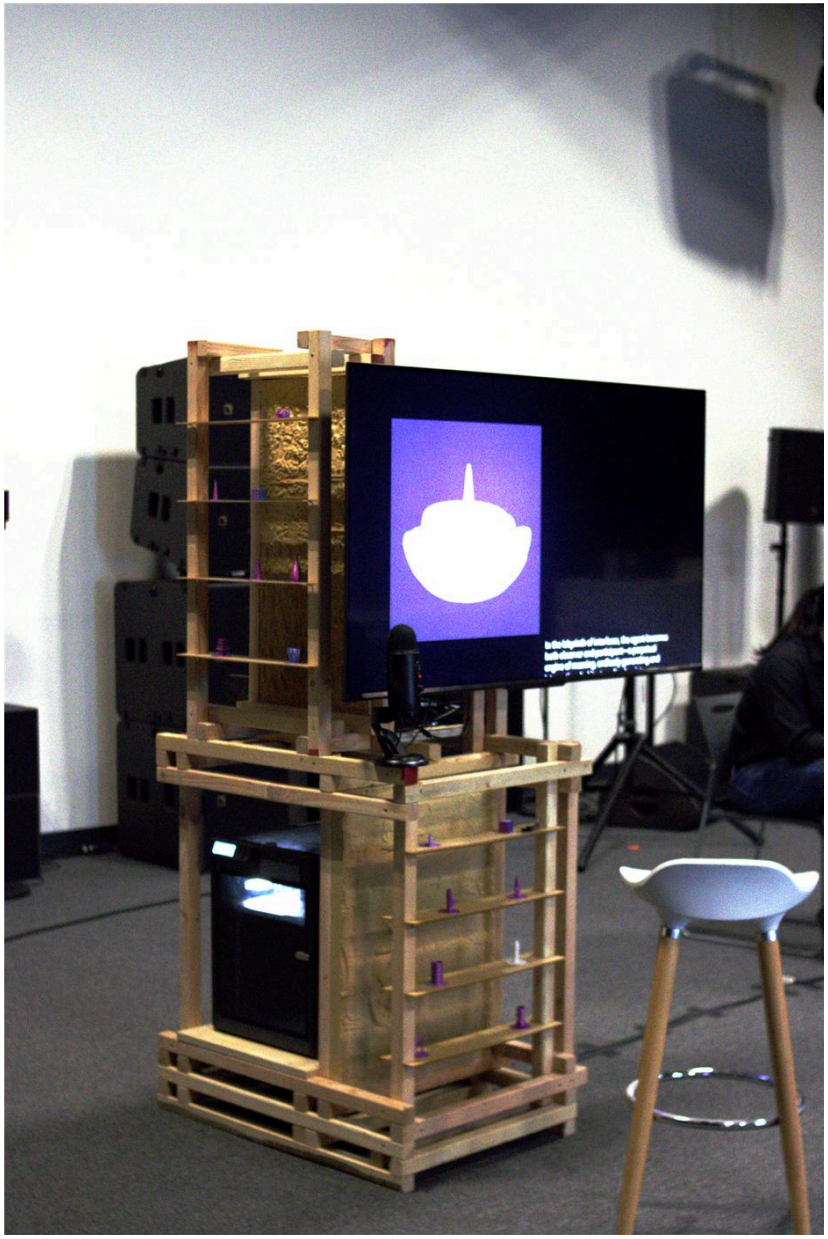


BURNT WOOD MEMORY ARTIFACT, STUDY 5



BURNT WOOD MEMORY ARTIFACT, STUDY 6

OBJECT STUDIES



FINAL INTERACTIVE INSTALLATION WITH SCREEN AND SPEAKER RIG



04

DESIGN SCHOOL IN MEXICO CITY

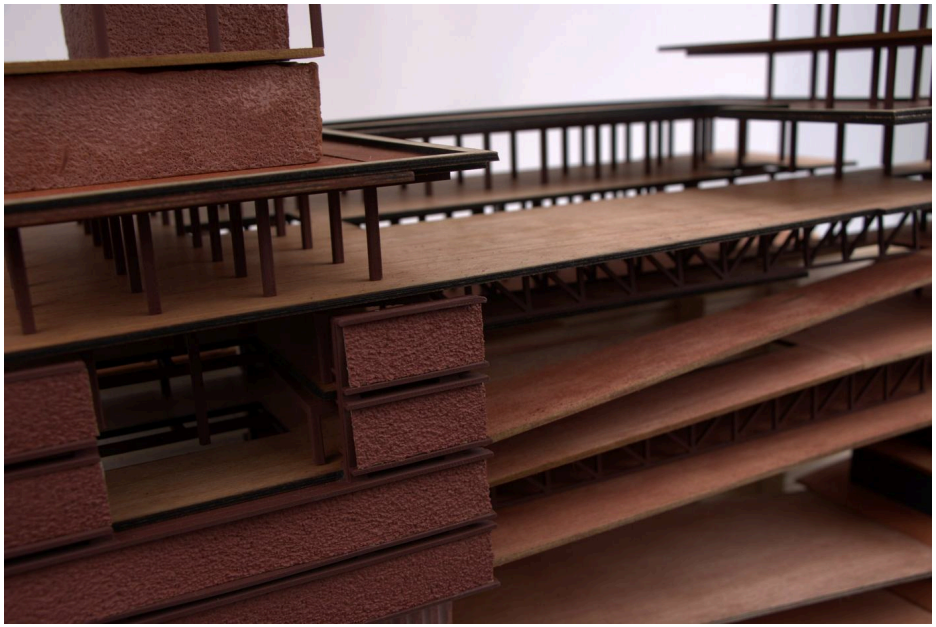
SP25, 2GB STUDIO, WITH NATE SMITH
ADVISOR · ZEINA KOREITEM

An addition to the IMSS headquarters, Mexico's social security bureau, in Mexico City: a design school that plugs into the existing office building without major changes to its structure. On our studio trip we read the city as a palimpsest, built in layers, with new buildings and materials laid directly against the ones before them. The proposal continues that reading. Studios, classrooms, a gallery, a café and a library stack as blocks on one another and on the IMSS building, a public plaza opens at street level, and a mesh facade drapes over the whole: one more layer of the city.

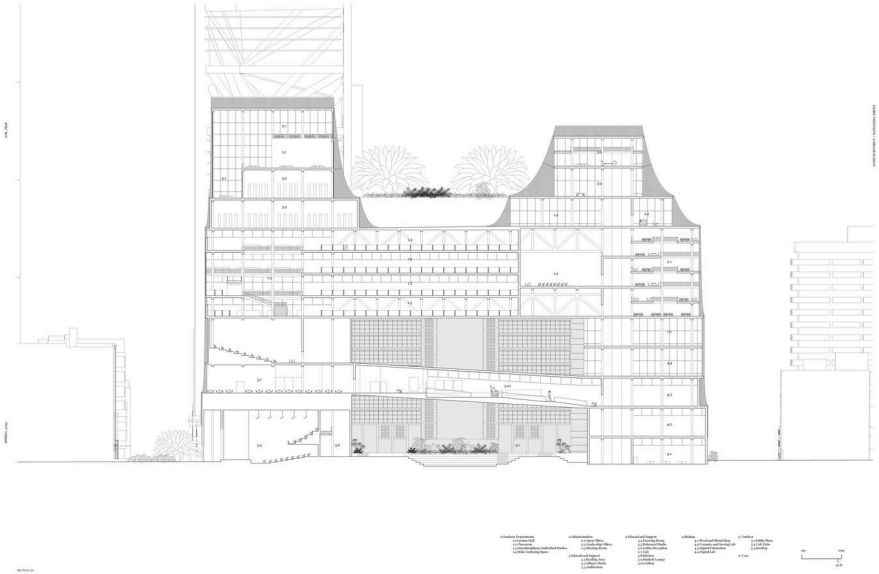
*Made with Nate Smith



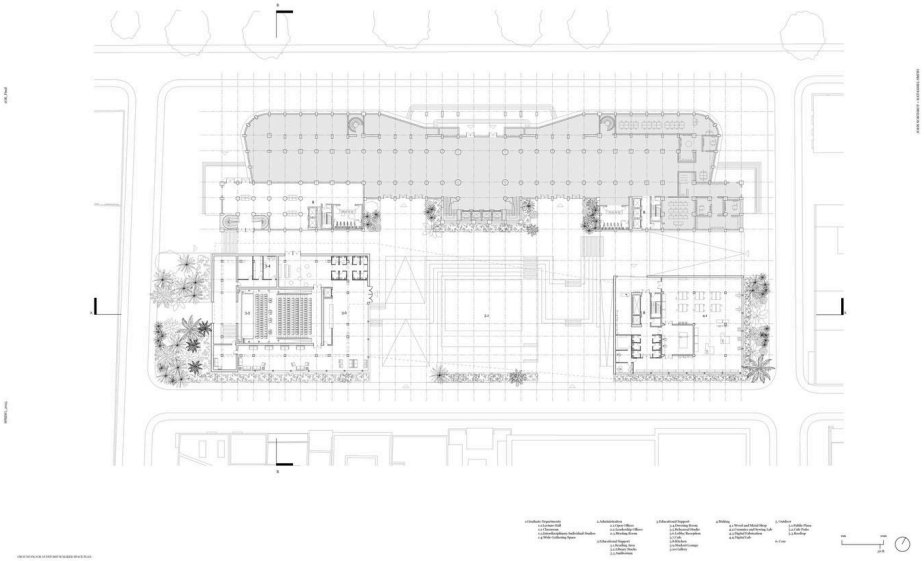
MODEL | FINAL MODEL WITH THE DRAPED MESH FACADE



MODEL | FLOOR PLATES, RAMPS AND TRUSSES



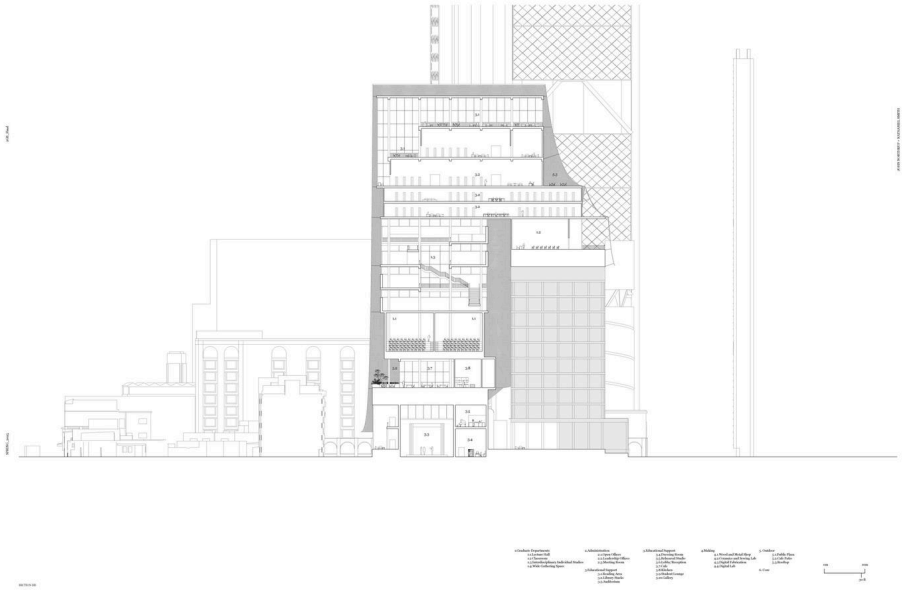
DRAWING | SECTION AA



DRAWING | GROUND FLOOR PLAN AND PUBLIC PLAZA



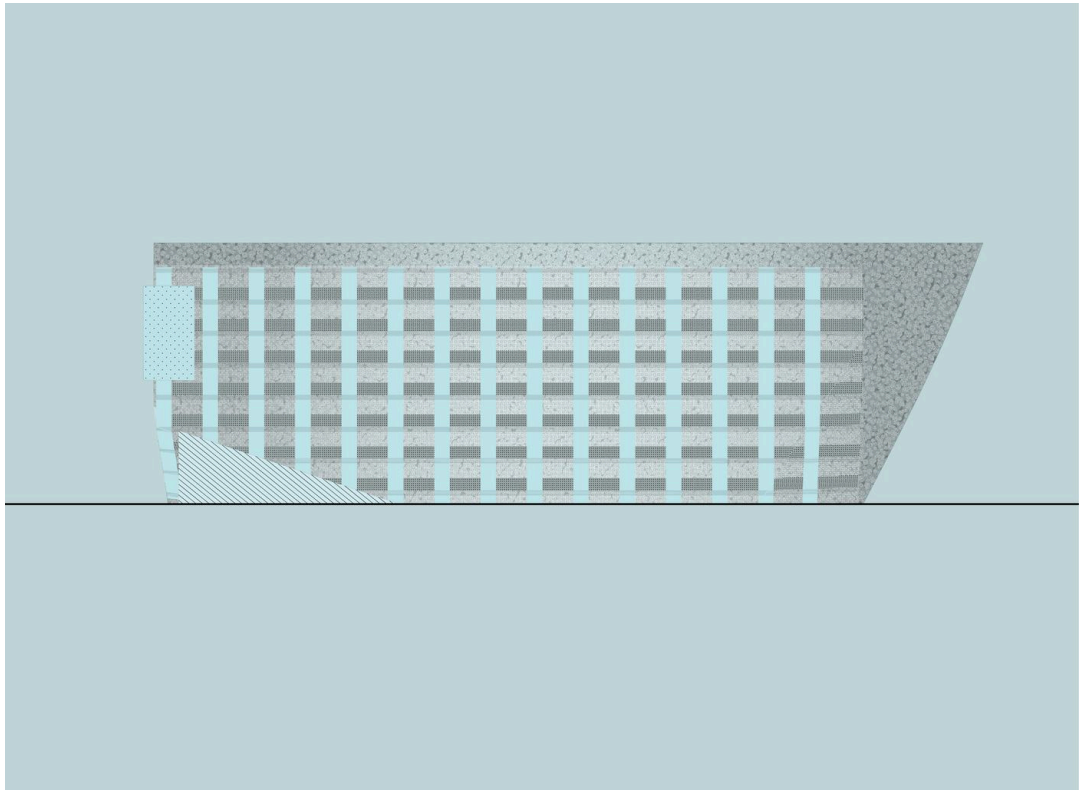
MODEL | BRICK-CLAD BLOCKS AND TRUSS BRIDGE



DRAWING | SECTION BB



MODEL | STUDY MASSING IN THE CITY SITE MODEL



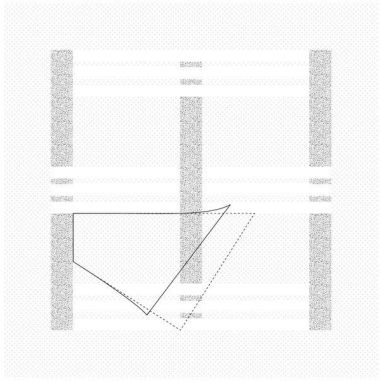
05

LITTLE TOKYO THEATER

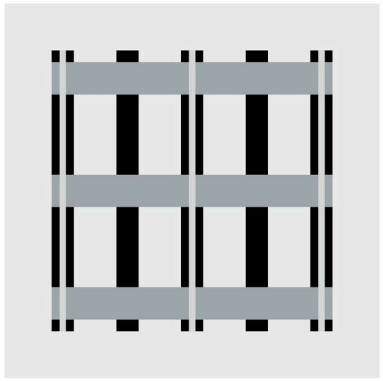
FA23, DS 1100, 1GA STUDIO
ADVISOR · MATTHEW AU

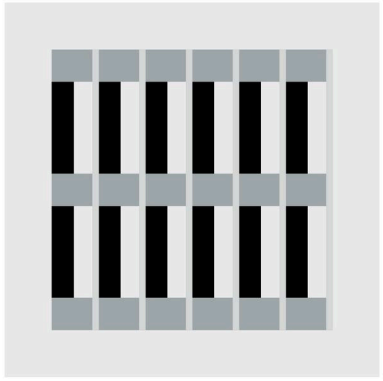
This first semester studio project stems from a study of grid systems, materiality, galleries, and theatres. The initial measurement grids were carefully constructed, using the limitations of the tape itself and the gap in named measurements vs. actual measurements. The grids operate in a way that looks at material falsities, as well as transparency. Through these measurement systems, they are then expanded into large boards. These boards are then used to produce a massing model. Ultimately realized as a performing arts theater in the Little Tokyo District, the Measurement Grid is transformed into a space that offers a place for community to gather in an unconventional theater setting where actors and patrons are active participants in the shows on display.

GRID STUDIES

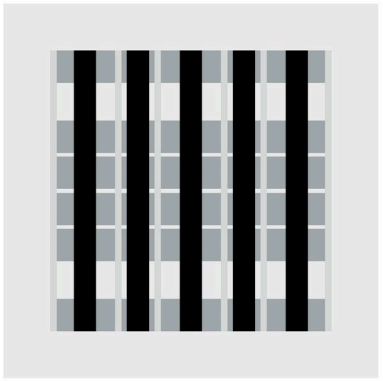


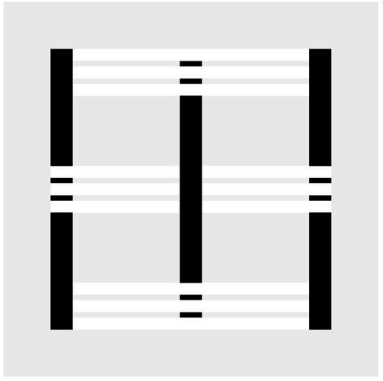
MEASUREMENT GRID DIAGRAM WITH TETRAHEDRON





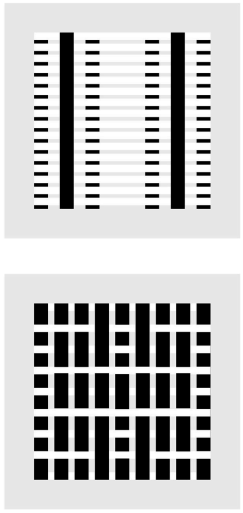
DIGITAL GRID FIGURE STUDY 1 AND 2



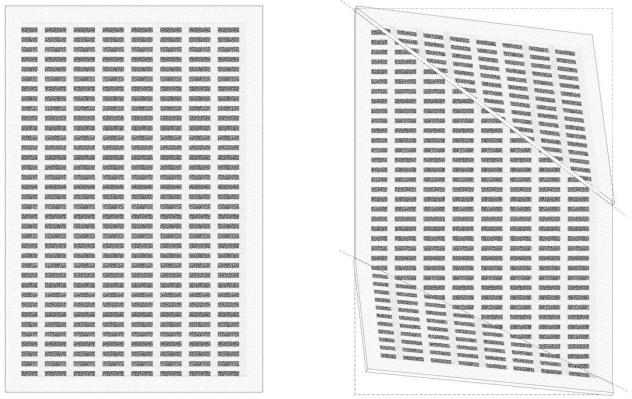


DIGITAL GRID FIGURE STUDY 3 AND 4

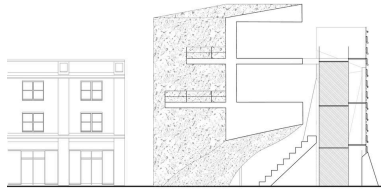
GRID STUDIES



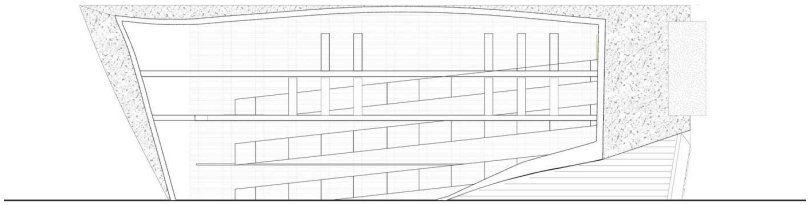
DIGITAL GRID FIGURE STUDY 5 AND 6



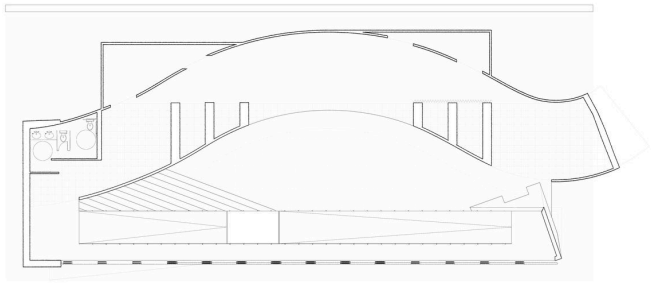
DIGITAL GRID STUDY, PRINTED BOARD FLAT AND BENT



GRID STUDY DRAWING, SHORT SECTION

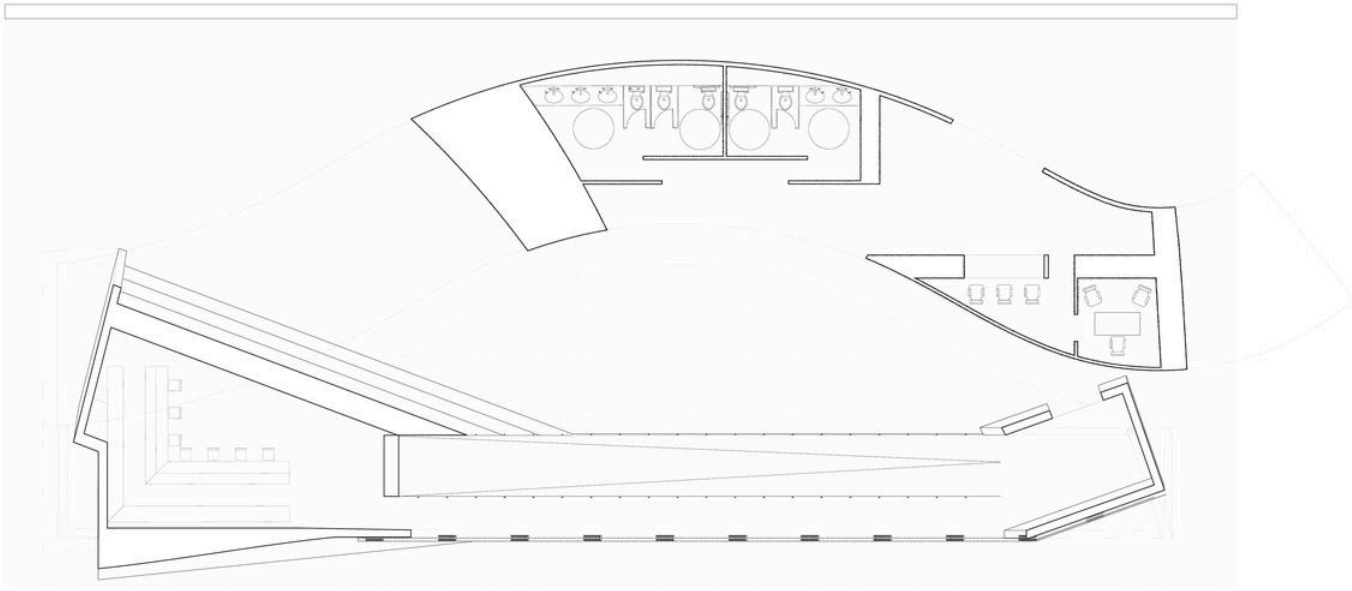


GRID STUDY DRAWING, LONG SECTION



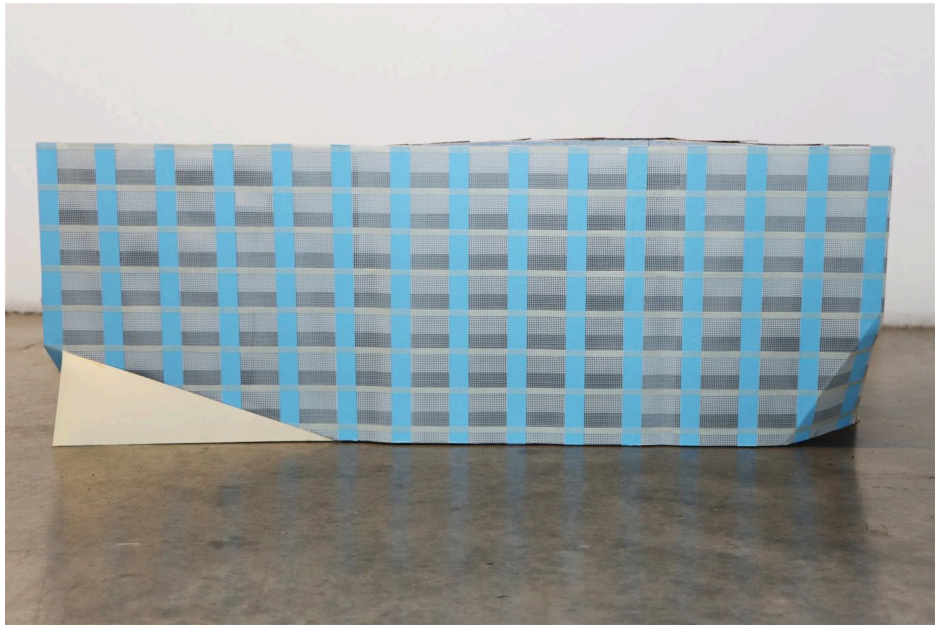
GRID STUDY DRAWING, SECTION

GRID STUDIES

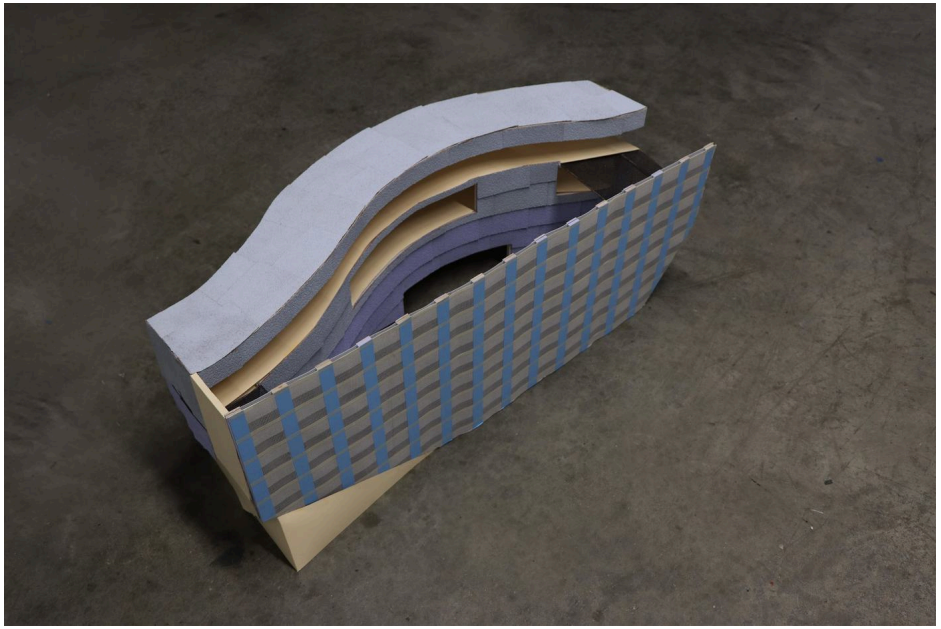


GRID STUDY DRAWING, PLAN

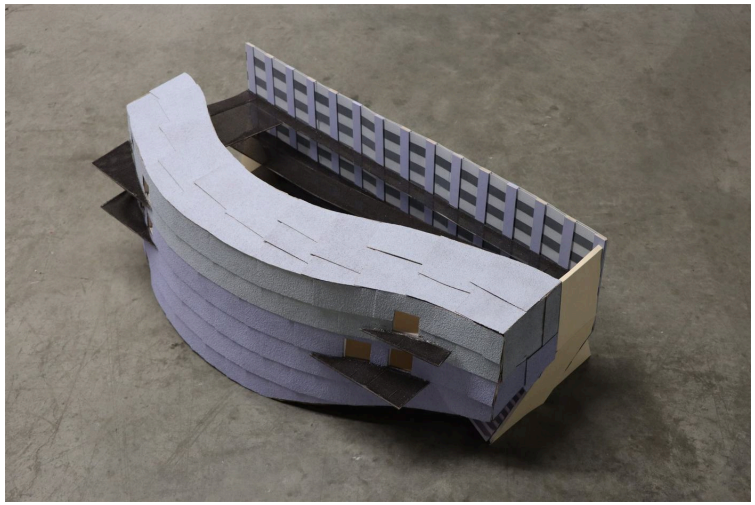
FINAL MODEL



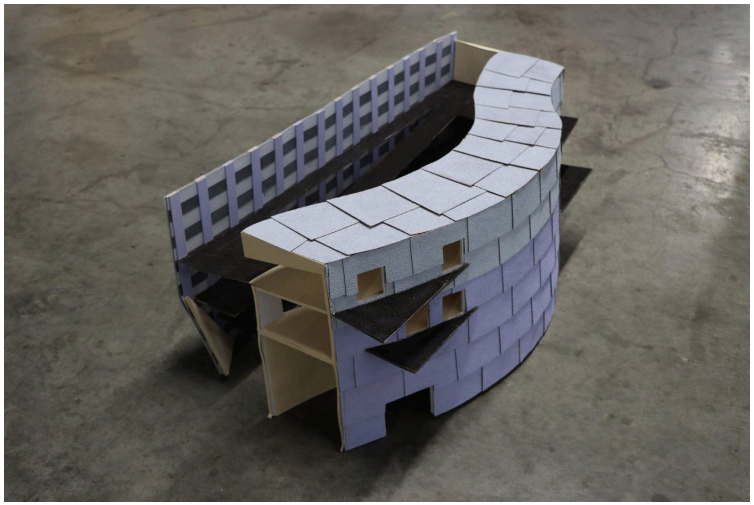
FINAL MASSING MODEL, PRIMARY VIEW



FINAL MASSING MODEL, VIEW 2



FINAL MASSING MODEL, VIEW 3

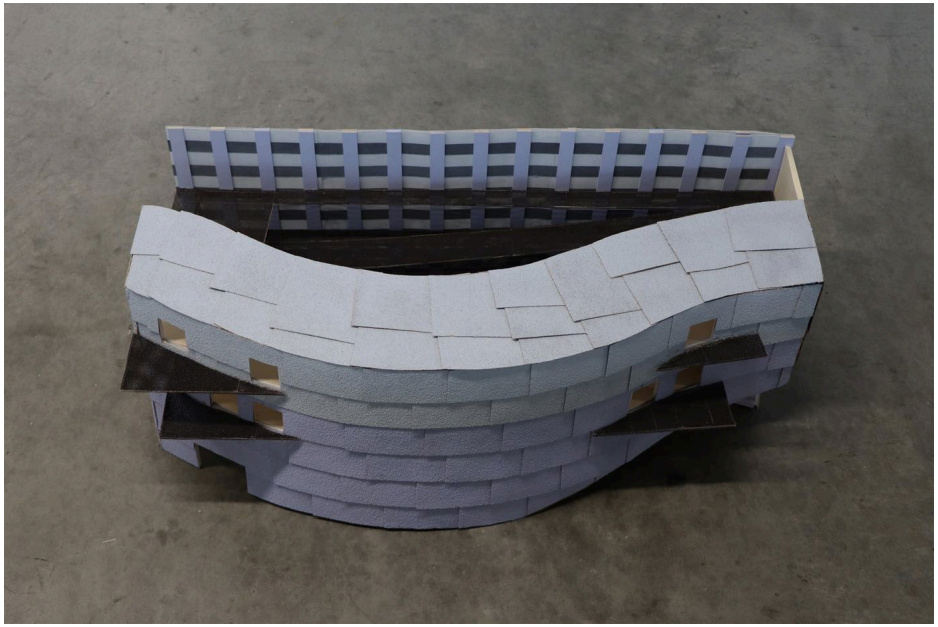


FINAL MASSING MODEL, VIEW 4

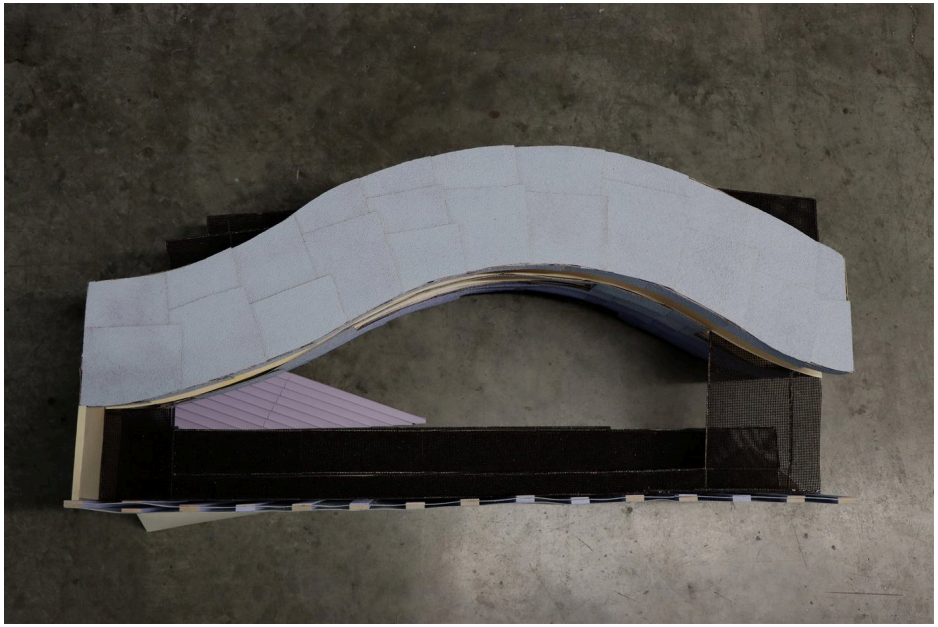


FINAL MASSING MODEL, VIEW 5

FINAL MODEL

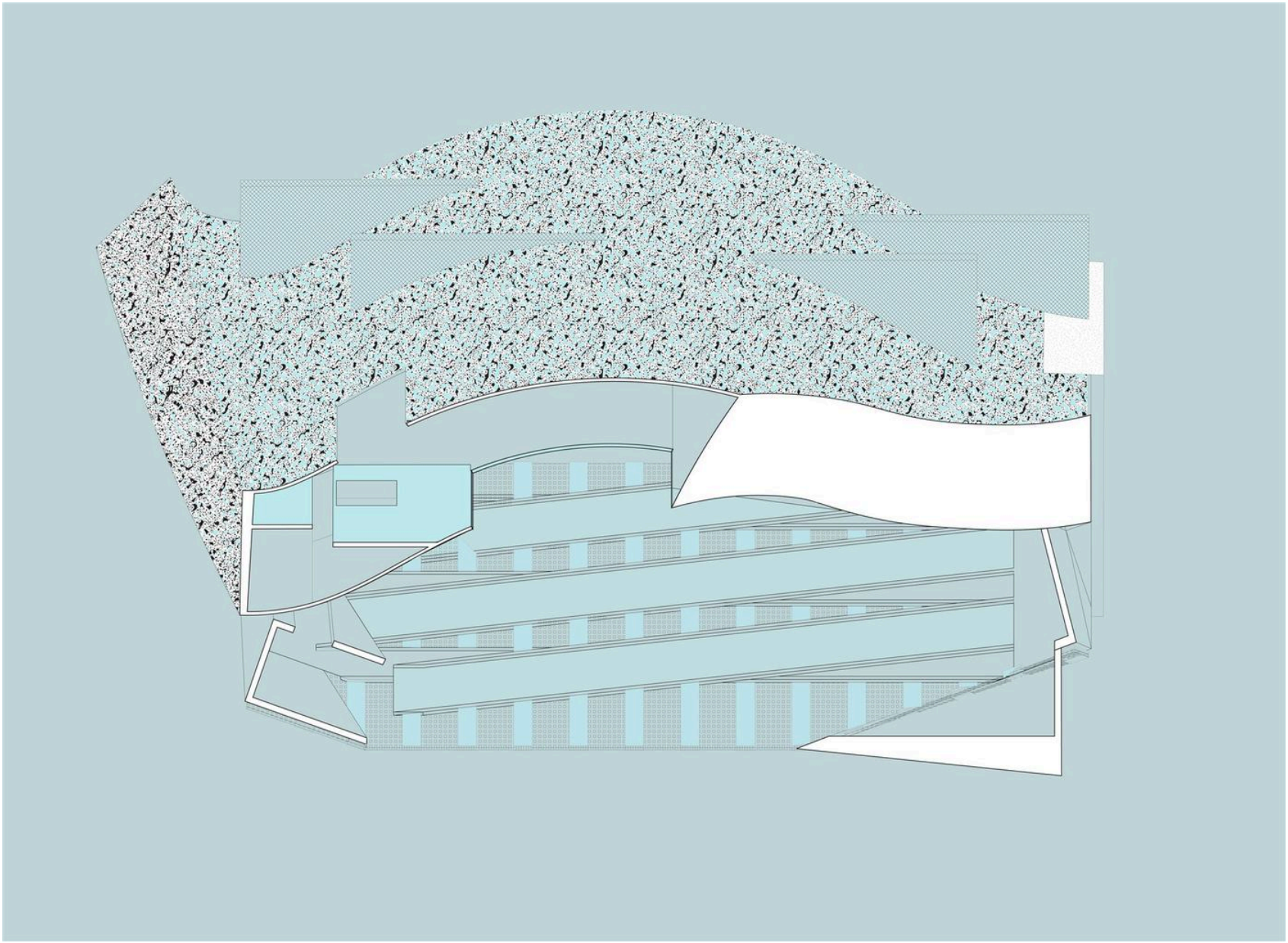


FINAL MASSING MODEL, ELEVATION VIEW

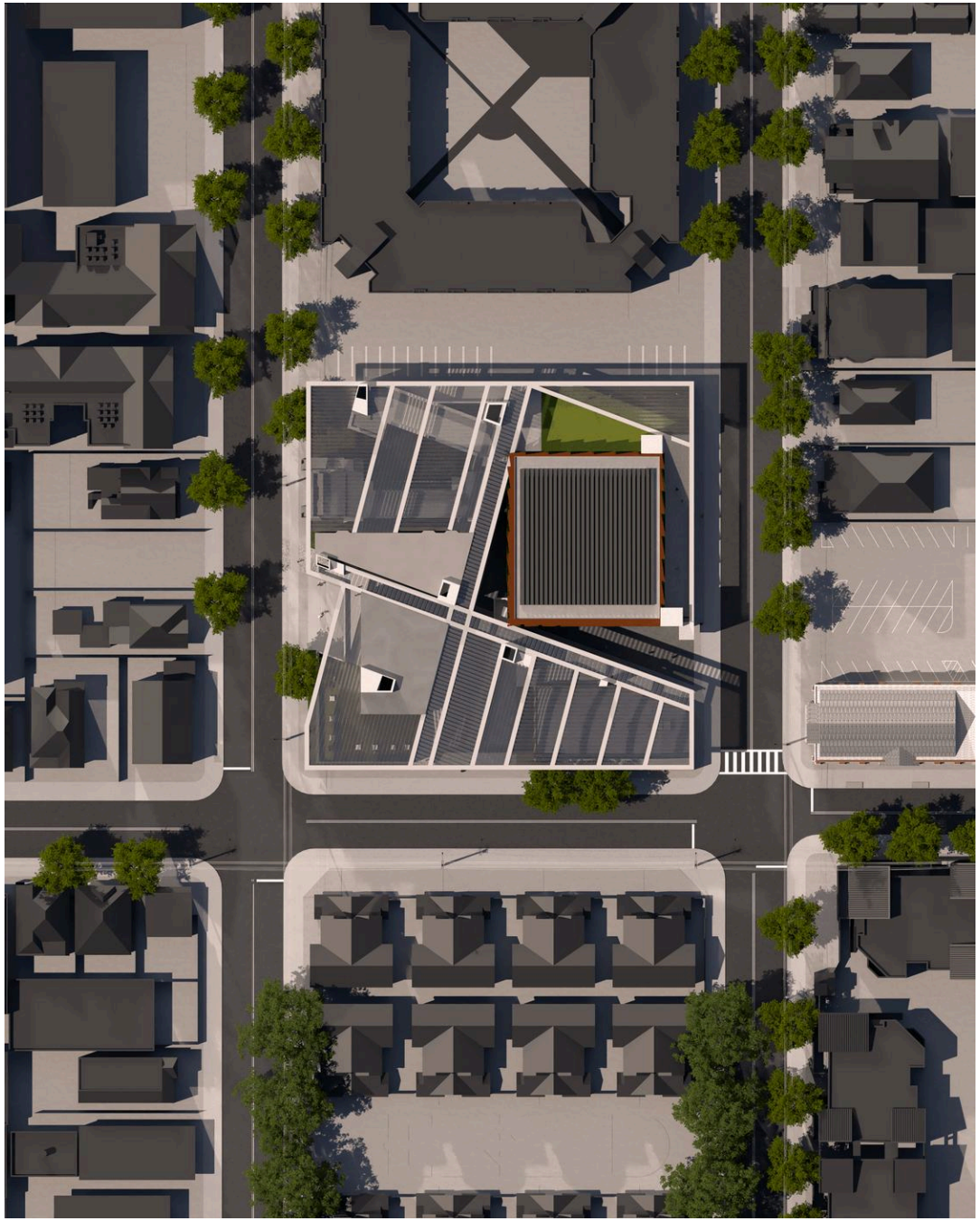


FINAL MASSING MODEL, VIEW 7

FINAL MODEL



FINAL MODEL COMPOSITE SECTION DRAWING



06

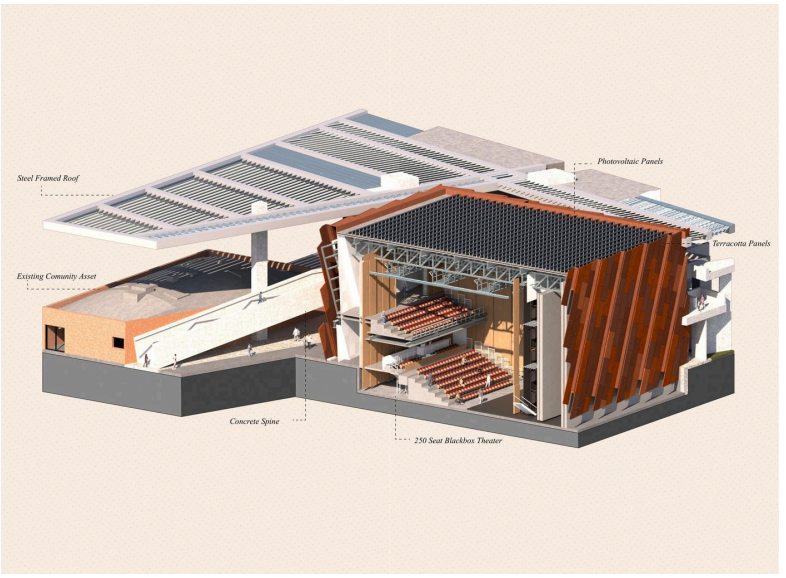
LA DANCE SHED

FA24, 2GA STUDIO, WITH MILAN SLEDGE
ADVISOR · MARCELO SPINA

This project for the Los Angeles Dance Project (LADP) seeks to bring the community of Pico-Union together on a campus that serves and connects the community. Located directly across from the Pico-Union Project, the site was surgically opened up in order to connect via sightlines and circulation to this important community asset. A unique goal among this project was to preserve many of the existing structures on the site and repurposing the materials from buildings that were to be removed. With a raised circulatory system, visitors are encouraged to meander throughout the site, engaging with it from within the in-between spaces. The roof serves as not only shade for warm summer days, but also acts as an organizing device for the discrete elements of the site, bringing them together in a collective, just as is intended with the surrounding community.



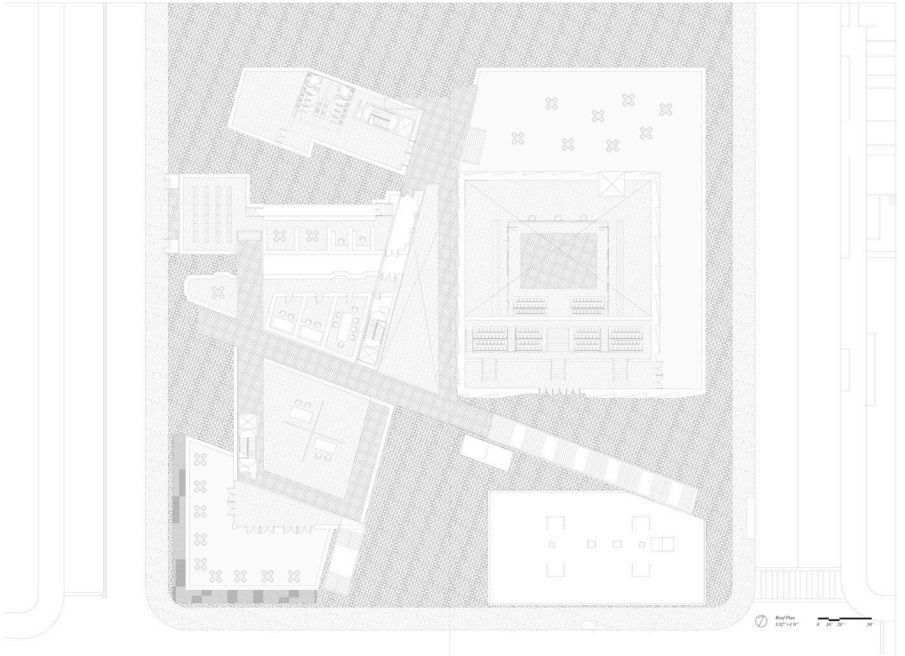
RENDER | FROM THE PICO-UNION SIDE OF THE SITE



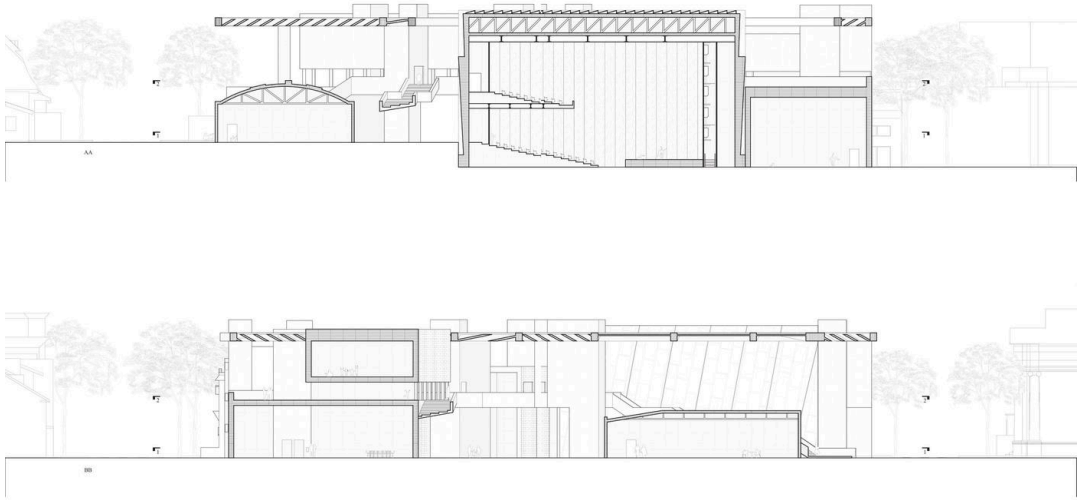
AXONOMETRIC | STEEL-FRAMED ROOF, CONCRETE SPINE AND 250-SEAT BLACK BOX TH...



PLAN | LEVEL 1



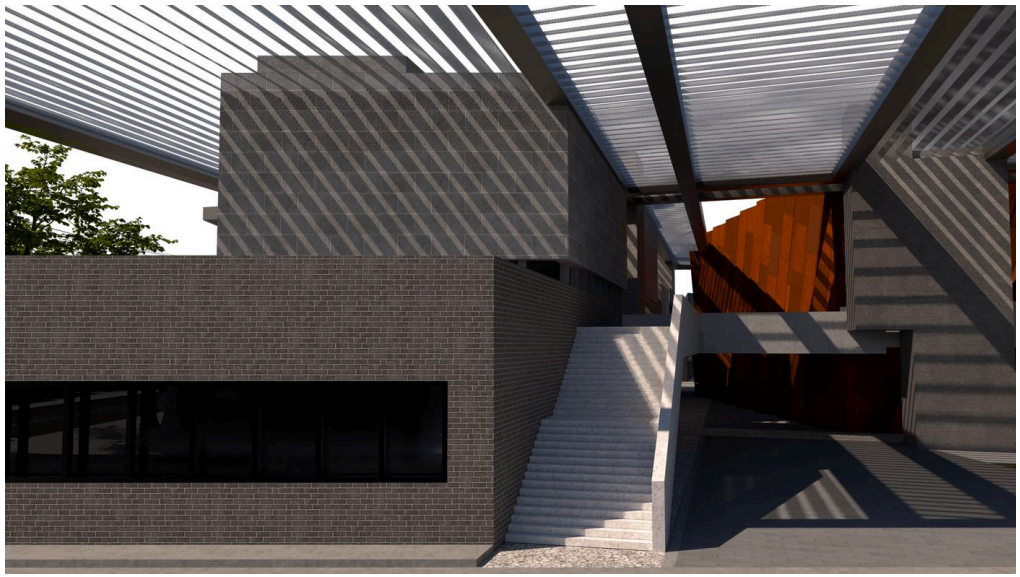
PLAN | LEVEL 2



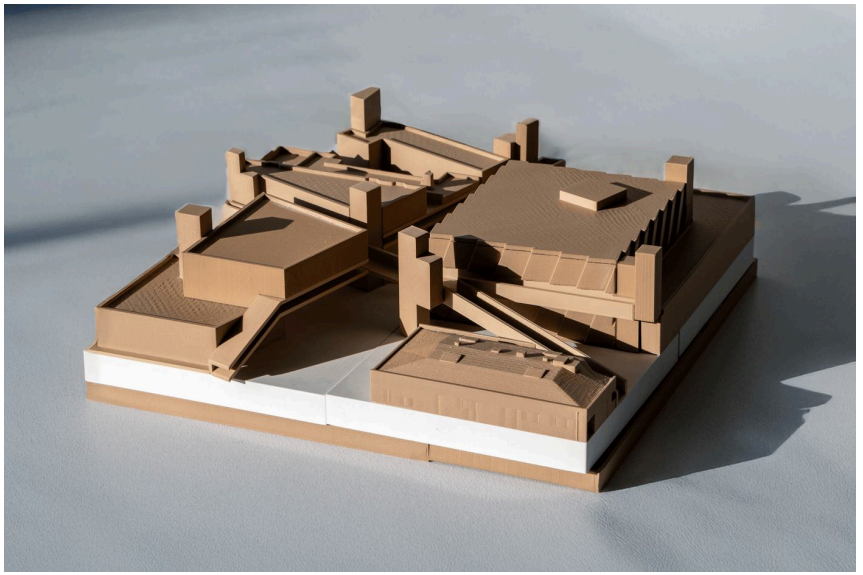
SECTIONS | THROUGH THE THEATER AND SPINE



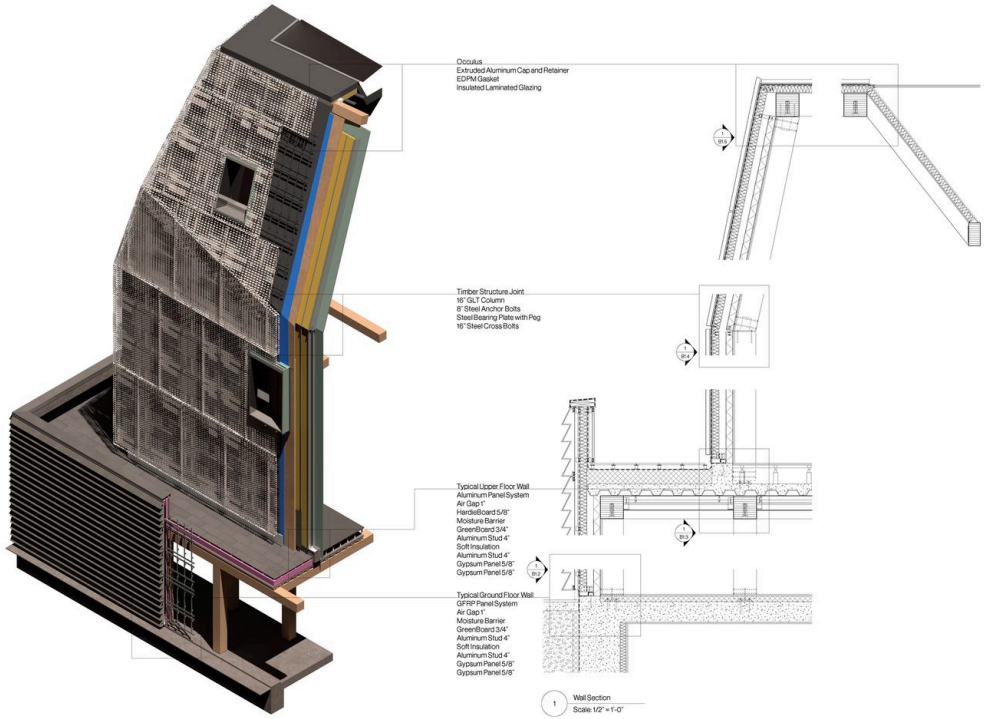
ELEVATIONS | STREET FRONTAGES, TERRACOTTA PANEL FACADE



RENDER | TOWARD THE CAMPUS CAFE



MODEL | FINAL MODEL

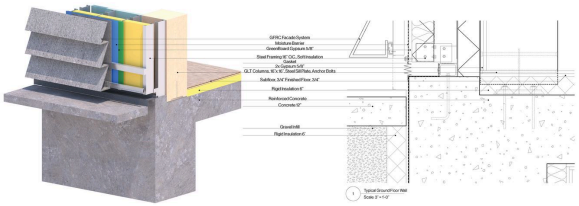


07

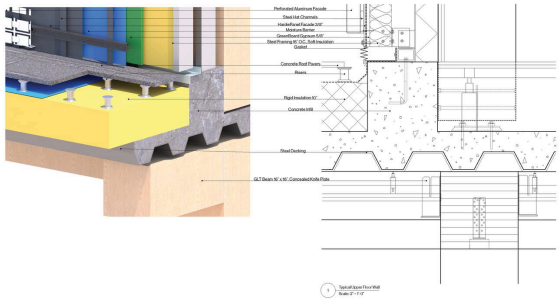
DESIGN DOCUMENTS

DESIGN DOCUMENTS
ADVISOR · HERWIG BAUMGARTNER

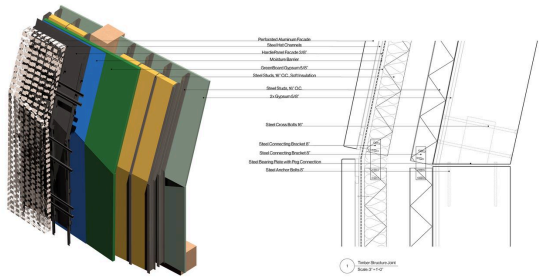
This course studies the implementation of design, technology, and an understanding of building systems. Students are tasked with preparing Design Document drawings from a previous year's studio project and are asked to bring it to life with real-world materials, budgets, and environmental factors.



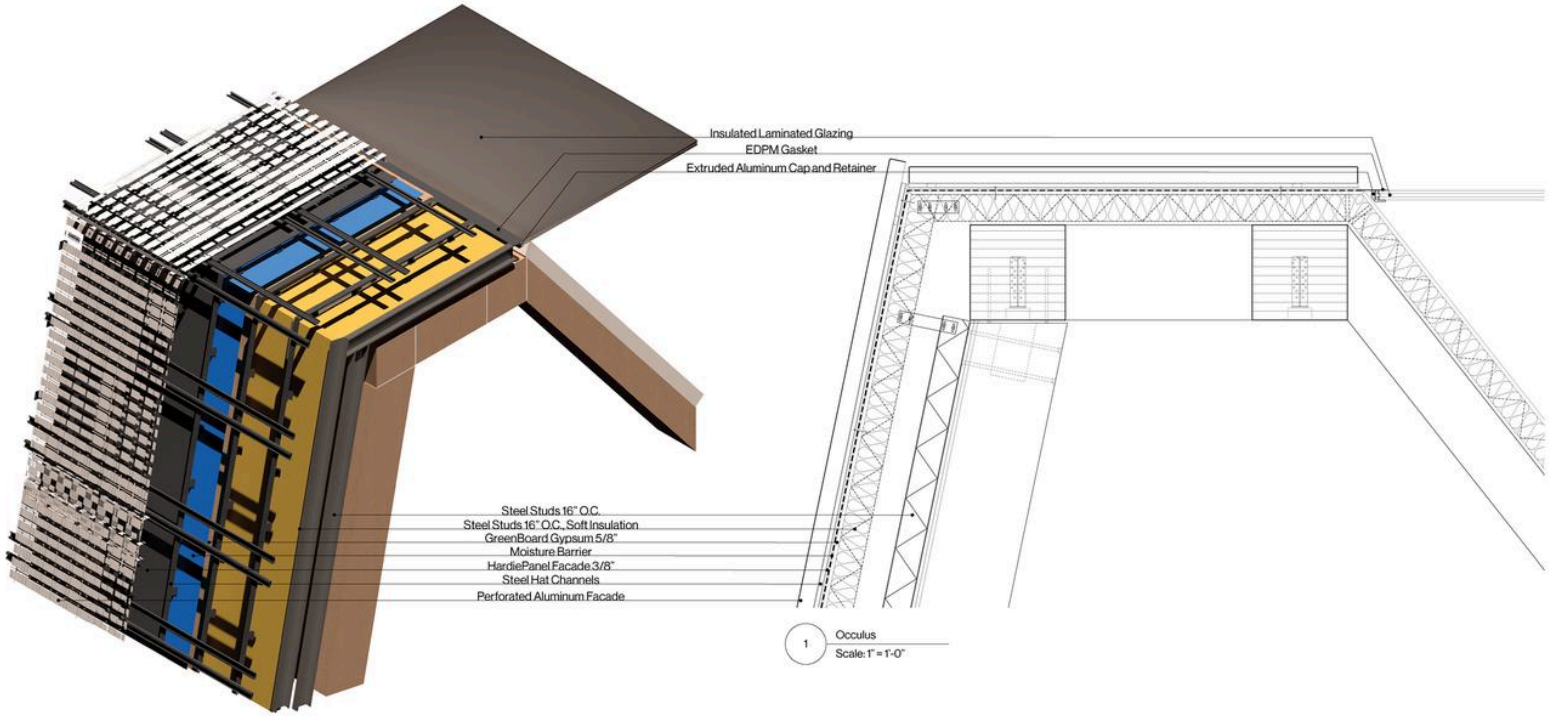
TYPICAL GROUND FLOOR WALL DETAIL, EXPLODED AXONOMETRIC ...



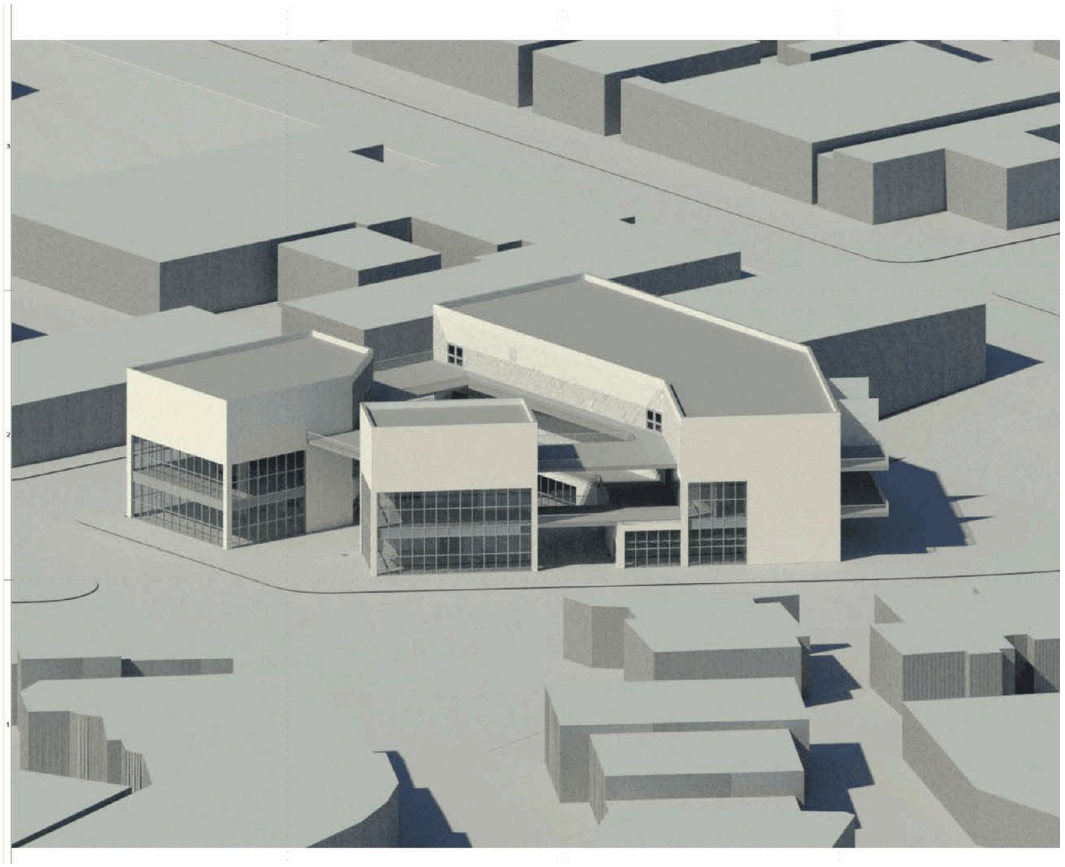
TYPICAL UPPER FLOOR WALL DETAIL, EXPLODED AXONOMETRIC AN...



TIMBER STRUCTURE JOINT DETAIL, EXPLODED AXONOMETRIC AND S...



OCULUS SKYLIGHT DETAIL, EXPLODED AXONOMETRIC AND SECTION



08

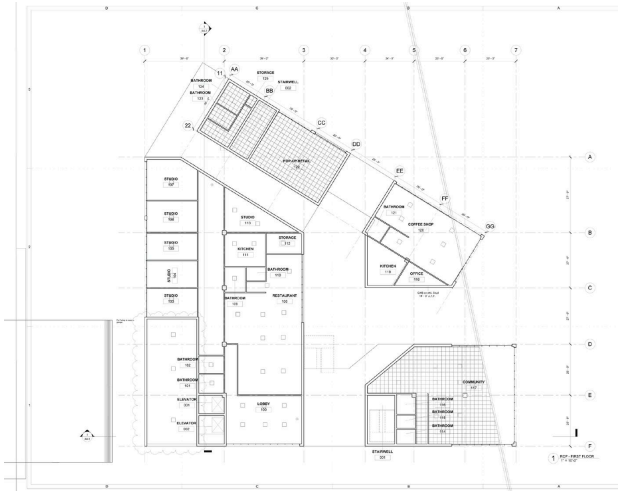
ADVANCED PROJECT DELIVERY

ADVANCED PROJECT DELIVERY
ADVISOR · PAVEL GETOV AND KARENZA HARRIS

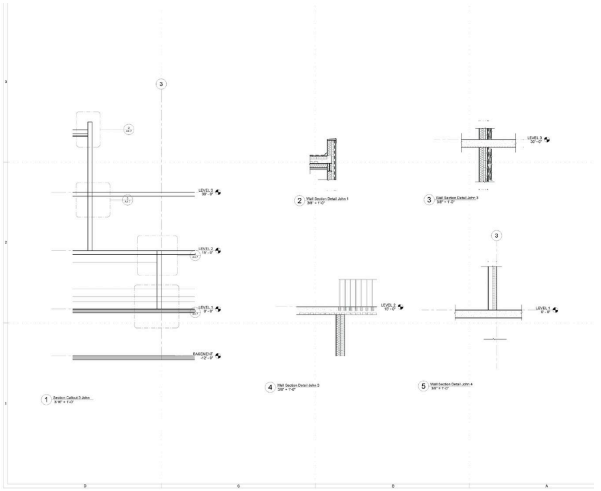
A Construction Documents course focused on taking a design through to buildable detail — resolving material assemblies, structural connections, and building systems into a coordinated drawing set. Site plans, reflected ceiling plans, and wall section sheets were developed to real-world standards of dimension, specification, and constructability.



SITE PLAN SHEET



REFLECTED CEILING PLAN, FIRST FLOOR



DETAILS AND WALL SECTION SHEET



09

ELF-O-MATIC-3000

FA22, MONSTER CITY STUDIOS
ROLE · TECHNICAL DESIGNER & PROJECT MANAGER

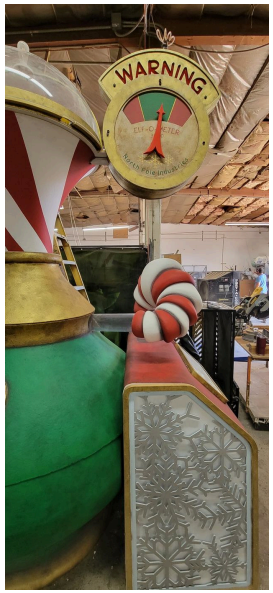
A holiday show piece built at Monster City Studios: a steam-powered candy machine with a neon marquee, a lit viewing window of tumbling balls, a STEAM grille, a pressure gauge and a candy-striped dome. Behind the theming is a working show, with lighting, moving parts and an operator panel to reset it between guests.



BUILT WORK | THE MACHINE RUNNING IN THE SHOP (VIDEO: JCNORTHRUP.COM/PROJECTS/E...



BUILT WORK | MARQUEE, GAUGES AND DOME



DETAIL | WARNING GAUGE, ...



DETAIL | OPERATOR PANEL: ...



DETAIL | SERVICE HATCH WITH LABELED CONTROL SWITCHES



10

DETAILS, DETAILS

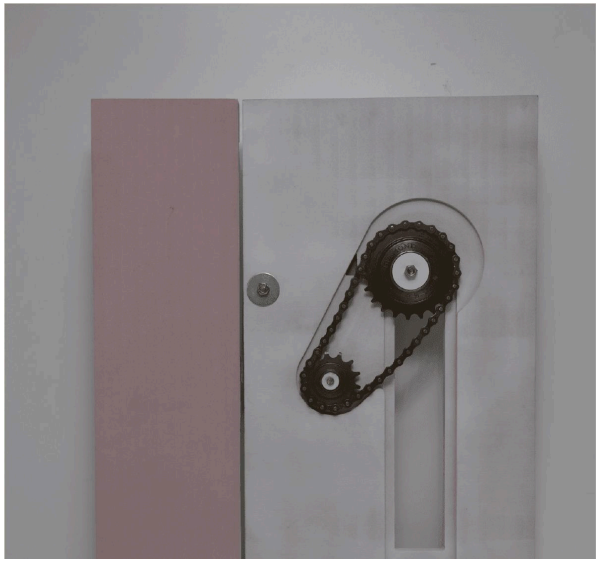
F25, DETAILS, DETAILS
ADVISOR · DWAYNE OYLER

A doorlock mechanism inspired by Frank Lloyd Wright's detail work, specifically the door mechanisms in the Hollyhock House.

*All work made with partners Chia-Chen Li and Baizhen Yu



FOAM PROTOTYPING PROCESS



PROTOTYPE INTERIOR MECHANISM



FINAL DOOR MECHANISM, CLOSE-UP